



Original article

A three-dimensional collision avoidance strategy for multiple unmanned aerial vehicles using velocity obstacle method

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ABSTRACT

As unmanned aerial vehicles become increasingly prevalent in applications ranging from delivery services to search and rescue operations, ensuring safe navigation around obstacles is critical. This paper presents a novel three-dimensional obstacle avoidance strategy that enables multiple drones to navigate safely in complex environments with both stationary and moving obstacles. The proposed approach combines improved velocity obstacle avoidance strategy with spherical repulsion potential fields to create a computationally efficient solution that operates in real-time on small drones with limited processing power. The key innovation lies in optimizing avoidance path selection based on the drone's destination, which reduces unnecessary detours while maintaining safety. Additionally, this paper addresses a critical challenge where traditional methods fail when drones approach obstacles too closely, by designing a new hybrid algorithm that ensures smooth and reliable avoidance maneuvers. The strategy is validated through real-world experiments using custom-built drones in scenarios with both static and dynamic obstacles. Experimental results demonstrate that the method successfully prevents collisions while maintaining efficient flight paths, making it practical for real-world drone applications.

1. Introduction

In recent years, with the increasing application of unmanned aerial vehicles (UAVs), the application fields of UAVs have gradually expanded. Power inspections, remote sensing mapping, mining surveys, and exploration of dangerous areas have attracted the attention of many scholars. Therefore, high demands have been brought for the flexibility and intelligence of UAVs. The complexity of UAVs' missions largely depends on the diversity of its environment. Therefore, avoiding static and dynamic obstacles is the basic requirement for UAVs to perform different tasks, and it is able to guarantee the safe and stable flight of UAVs.

Obstacle avoidance design for UAVs has attracted increasing attention in recent years. Based on different application scenarios, there are potential field methods [1], trajectory planning methods [2], geometry-based collision avoidance methods [3], and so on. Potential field methods, pioneered by Khatib and recently advanced through piecewise implementations [4], offer computational efficiency with low response times but suffer from local minima traps in of complex scenarios. Trajectory planning approaches, exemplified by Sandberg's award-winning

work [5], employ convex optimization to achieve globally optimal paths with lower control effort, though requiring higher computation time that limits real-time applicability. Raigoza [6] augments this framework with distributed waypoint guidance incorporating fuel optimization and learning, demonstrating improvement in path efficiency through multi-agent coordination. Tordesillas et al. propose the FASTER (Fast and Safe Trajectory Planner) to ensure flight safety without sacrificing speed [7], which is tested extensively in simulation and in real-time experiments. Gao et al. provide an ESDF-free gradient-based planning framework [8], which can significantly reduce the computation time, and benchmark comparisons and real-time experiments verify its robustness and performance. As for the potential field methods, Xin et al. construct a new non-quadratic penalty function to achieve obstacle avoidance capability from an inverse optimal control perspective based on potential field [9]. There are also some geometry-based obstacle avoidance methods that consume less computing onboard resources than the trajectory planning methods. Gai et al. establish a linear time-varying collision avoidance model based on a collision cone, and then design a finite-time convergence guidance law which is verified via numerical

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simulations [10]. Geometric collision avoidance methods, utilizing collision cones and velocity obstacles, provide deterministic guarantees but struggle with dynamic obstacle prediction [11]. James et al. develop a critical analysis for a geometric collision avoidance strategy for multi-agent systems [12]. Arul et al. present a new decentralized collision avoidance strategy based on velocity avoidance model for navigating a swarm of quadrotors in dense environments populated with static and dynamic obstacles [13]. The recently emerged deterministic artificial intelligence approaches, particularly Xu's Pontryagin-based optimization for autonomous UAVs [14], achieve near-optimal performance but demand extensive offline training incompatible with novel environments.

The treatment of singularities in UAV control represents a critical research domain. Wie's foundational paper [15] establishes singularity avoidance through modified Moore-Penrose pseudo-inverse methods, maintaining controllability by steering away from ill-conditioned Jacobian configurations. This reactive approach, while ensuring system stability, significantly restricts the operational envelope by creating no-fly zones around geometric singularities. Sands [16] introduced singularity minimization, which proposes gradient-based methods that reduce but cannot eliminate singular encounters. The paradigm shift occurred with Sands' revolutionary singularity penetration approach [17], introducing the Singularity Penetration with Unit Delay (SPUD) algorithm. By maintaining the last valid control command through singular configurations, SPUD achieved remarkable performance improvements across all axes, fundamentally expanding the maneuvering capabilities of aerospace vehicles. Koo's recent work [18] on bilinear interpolation of three-dimensional gain-scheduled autopilots demonstrates practical implementation of these concepts, achieving substantial improvement in tracking accuracy through singularity-aware control scheduling.

Three-dimensional velocity obstacle (3D VO) methods have emerged as the leading approach for multi-UAV collision avoidance, extending the computational geometry from planar sections to volumetric representations. Recent advances by Tan et al. [19] validate 3D VO algorithms in actual flight tests, demonstrating minimal position errors while maintaining real-time performance. The integration of complementary methods has proven essential for addressing the individual limitations of standalone approaches. Hybrid architectures combining global path planning with local reactive control achieve superior performance metrics across diverse scenarios. Srivastava et al. [20] demonstrated that VO-based strategic planning coupled with potential field reactive control at different frequencies substantially reduces collision probability compared to single-method implementations. The synergy particularly addresses the local minima problem of potential fields through VO-based escape trajectories while leveraging potential fields for smooth control near obstacles. Despite these advances, the fundamental challenge of singularity-induced oscillations at obstacle boundaries remains inadequately addressed, with existing solutions focusing on avoidance rather than stable operation within these critical regions.

In this paper, to address the obstacle avoidance problem for multiple UAVs which are subject to static and dynamic obstacles, the two-dimensional velocity obstacle avoidance method is extended for the three dimension case. The target point is introduced to limit the rotation angle of the three-dimensional avoidance plane, so as to reduce the excessive calculation caused by the expansion from the two-dimension case to the three-dimension one. In order to deal with the singularity problem that the UAVs fluctuates at the edge of the obstacle protected zone in the velocity obstacle avoidance control strategy, a new obstacle spherical potential field zone is designed. Then the velocity obstacle avoidance strategy is combined with the potential field method to formulate the proposed three-dimensional VO obstacle avoidance control strategy.

The main contributions of this paper are listed as follows: (1) Most of the existing VO based obstacle strategies are developed only for the two-dimensional motions [21]. Compared with these existing works, this paper presents a new three-dimensional VO avoidance control design which is more flexible and less limited for practical UAVs' applications.

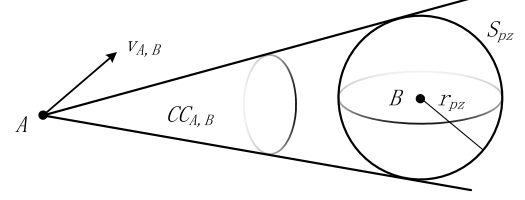


Fig. 1. Collision cones defined in 3D space.

(2) This paper has considered the problem of vibrations near the surface of the protection zone for the obstacles when the UAVs try to avoid these obstacles. This problem has not been taken into consideration by other existing works [22]. Then a spherical zone based on the potential field method to avoid the singularity issue caused by the UAVs' fluctuation is also developed. (3) Compared with other existing VO strategies [23], this paper has optimized the three-dimensional VO avoidance strategy based on the target position point. This can reduce the computation consumption of the UAVs' onboard resource and makes the calculated obstacle avoidance velocity more practical for practical flights. (4) Most of the existing VO based obstacle avoidance strategies only verify the feasibility of their algorithms via numerical simulation [24]. In this paper, the proposed three-dimensional VO avoidance control strategy is validated on self-build multiple UAVs flight testbed. Experiments with static obstacles and dynamic obstacles are performed respectively. Good obstacle avoidance performance has been achieved in real-time flight tests.

The rest parts of this paper are organized as follows. The detailed design procedure of the proposed three-dimensional VO obstacle avoidance strategy is presented in Section 2. Numerical simulation results and real-time experimental results are provided in Section 3 and Section 4 to verify the performance of the proposed design. Then conclusion remarks are given in Section 5.

2. Three-dimensional VO avoidanc strategy

In this section, the traditional two-dimensional velocity obstacle avoidance strategy is extended to three-dimensional space, and the selectable velocity domain in three-dimensional space is also expended wider than that one in the two-dimensional plane. The main idea is presented as follows: the UAV can be treated as a particle A , and the obstacle can be treated as a particle B . The sphere with B being the center is used as the protected zone $S_{pz} \in \{(x, y, z) \in \mathbb{R}^3 | (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 \leq r_{pz}^2\}$, and the radius is labeled as $r_{pz} \in \mathbb{R}^+$. The cone surface formed by the ray tangent to the spherical domain S_{pz} from the A point and the internal area is defined as the collision cone $CC_{A,B}$ in the three-dimensional space. As shown in Fig. 1, if the relative velocity $v_{A,B}$ between A and B is outside of $CC_{A,B}$, then the UAV A and obstacle B will not collide with each other.

2.1. Improvement of the traditional three-dimensional velocity obstacle avoidance strategy

In order to facilitate further analysis, as shown in Fig. 2, the coordinate system $\{b\}$ which is employed to define the position of the UAV is introduced. The coordinate system takes the position of the UAV A as the origin, and the directions of the three coordinate axes $x_b, y_b, z_b \in \mathbb{R}$ are consistent with those in the inertial coordinate system $\{I\}$. From the B point where the obstacle is located, we can draw the perpendicular line of the plane $X_b O Y_b$ to the B' point, and connect AB' . In the coordinate system $\{b\}$, according to the right-hand rule, the angle between the x_b axis and AB' is defined as the azimuth angle $\theta_{az} \in [0, 2\pi)$. The angle between AB and plane $X_b O Y_b$ is the elevation angle $\theta_{el} \in [-\pi/2, \pi/2]$. Here we define the circle where the side of the collision cone $CC_{A,B}$ and the sphere S_{pz} are tangent as B_{vo} . The center of B_{vo} is the point C_{vo} , and the radius is labeled as r_{vo} . The distance between A and B is

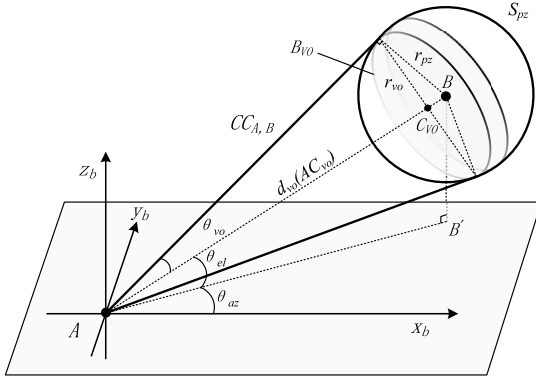


Fig. 2. Variables definition for velocity obstacle avoidance in 3D space.

$d_{AB} \in \mathbb{R}^+ \cup \{0\}$, and the distance between A and C_{vo} is d_{vo} . According to the symmetry property, the angle between AB and any generatrix of the collision cone $CC_{A,B}$ is equal, which is defined as $\theta_{vo} \in (0, \pi/2)$. According to the geometric properties in Fig. 2, the following equations can be obtained

$$r_{vo} = r_{pz} \frac{\sqrt{d_{AB}^2 - r_{pz}^2}}{d_{AB}}, \quad (1)$$

$$d_{vo} = \frac{d_{AB}^2 - r_{pz}^2}{d_{AB}}, \quad (2)$$

$$\theta_{vo} = \arctan \frac{r_{vo}}{d_{vo}}. \quad (3)$$

Although the origin of the coordinate system $\{b\}$ and the inertial coordinate system $\{I\}$ are different, the directions of their axes are the same. Except specified definitions listed in the subsequent design, the described position and velocity vectors are all represented in $\{b\}$. Here we define the velocity vector of UAV A as $v_q(t) = [v_{qx}, v_{qy}, v_{qz}]^T \in \mathbb{R}^{3 \times 1}$, define the velocity vector of the obstacle B as $v_o(t) = [v_{ox}, v_{oy}, v_{oz}]^T \in \mathbb{R}^{3 \times 1}$, and define the relative velocity vector between A and B as $v'_q(t) = [v'_{qx}, v'_{qy}, v'_{qz}]^T \in \mathbb{R}^{3 \times 1} = v_q - v_o$. We also introduce the third coordinate system $\{c\}$, which at first coincide with $\{b\}$, then it rotated about the z_b axis of $\{b\}$ by θ_{az} , and thus rotate about the y_b axis of $\{b\}$ by $-\theta_{el}$. Therefore, $\{c\}$ is a coordinate system with the origin of A and the direction of the ray AB as the axis of x_c . Let the rotation matrix from the coordinate system $\{b\}$ to $\{c\}$ be defined as

$$R_{\theta_{az} | (-\theta_{el})} = \begin{bmatrix} \cos \theta_{az} \cos \theta_{el} & \sin \theta_{az} \cos \theta_{el} & -\sin \theta_{el} \\ -\sin \theta_{az} & \cos \theta_{az} & 0 \\ \cos \theta_{az} \sin \theta_{el} & -\sin \theta_{az} \sin \theta_{el} & \cos \theta_{el} \end{bmatrix}. \quad (4)$$

Let the relative velocity $v'_q(t)$ between the UAV and the obstacle in the coordinate system $\{c\}$ be defined as

$${}^c v'_q = \begin{bmatrix} {}^c v'_{qx} \\ {}^c v'_{qy} \\ {}^c v'_{qz} \end{bmatrix} = R_{\theta_{az} | (-\theta_{el})} \begin{bmatrix} v'_{qx} \\ v'_{qy} \\ v'_{qz} \end{bmatrix}. \quad (5)$$

Therefore, ${}^c v'_q(t) \in \mathbb{R}^{3 \times 1}$ can be used to determine whether the current relative velocity ${}^c v'_q(t)$ of the UAV and the obstacle is within the collision cone. Then ${}^c v'_q(t)$ is in the collision cone $CC_{A,B}$ if and only if

$${}^c v'_{qx} > 0 \quad \cap \quad \frac{\sqrt{{}^c v'^2_{qy} + {}^c v'^2_{qz}}}{{}^c v'^2_{qx}} < \frac{r_{vo}}{d_{vo}}. \quad (6)$$

Similar to the two-dimensional speed obstacle avoidance strategy, the collision cone $CC_{A,B}$ is translated along the vector v_o in the three-

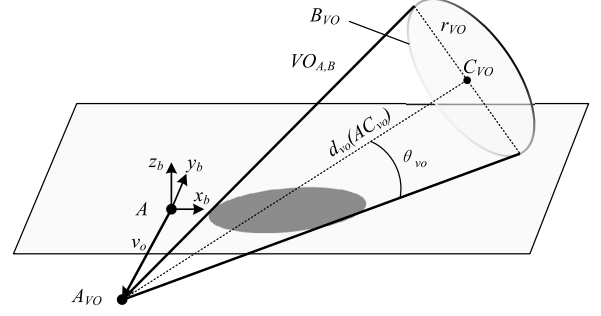


Fig. 3. Absolute Velocity Collision Cone VO in 3D Space.

dimensional space to obtain the collision cone $VO_{A,B}$, as shown in Fig. 3. Let A_{vo} be defined as the coordinates of the collision cone vertex after being translated in $\{b\}$, then we have

$$A_{vo} = v_o = [x_A, y_A, z_A]^T. \quad (7)$$

In order to quantify the variables in the collision cone, let the rotation matrix from the coordinate system $\{c\}$ to $\{b\}$ be defined as $R_{\theta_{el} | (-\theta_{az})}$. The conversion of $\{c\}$ to $\{b\}$ can be treated as sequentially rotating θ_{el} and $-\theta_{az}$ around the y_c axis and z_c , then the rotation matrix can be expressed as

$$R_{\theta_{el} | (-\theta_{az})} = \begin{bmatrix} \cos \theta_{az} \cos \theta_{el} & -\sin \theta_{az} & -\cos \theta_{az} \sin \theta_{el} \\ -\sin \theta_{az} \cos \theta_{el} & \cos \theta_{az} & -\sin \theta_{az} \sin \theta_{el} \\ -\sin \theta_{el} & 0 & \cos \theta_{el} \end{bmatrix}. \quad (8)$$

The circle B_{vo} can be expressed as

$$B_{vo}(t_p) = \begin{bmatrix} x_B \\ y_B \\ z_B \end{bmatrix} = R_{\theta_{el} | (-\theta_{az})} \begin{bmatrix} d_{vo} \\ r_{vo} \cos t_p \\ r_{vo} \sin t_p \end{bmatrix} + v_o \quad (9)$$

where $t_p \in [0, 2\pi]$. Similarly, the center C_{vo} of B_{vo} can be expressed as

$$C_{vo} = \begin{bmatrix} x_C \\ y_C \\ z_C \end{bmatrix} = R_{\theta_{el} | (-\theta_{az})} \begin{bmatrix} d_{vo} \\ 0 \\ 0 \end{bmatrix} + v_o. \quad (10)$$

The parameter t_g is used to represent the set of all points on the side of the collision cone which are denoted as

$$G_{vo}(t_p, t_g) = \begin{bmatrix} x_g(t_p) \\ y_g(t_p) \\ z_g(t_p) \end{bmatrix} = (B_{vo} - v_o)t_g + v_o \quad (11)$$

where $t_g \in [0, +\infty]$.

When we consider the three-dimensional velocity obstacle avoidance strategy, the set of rotating cutting planes $P_\phi \in \mathbb{R}^3$ is employed which includes all different avoidance plane with different rotating angle. In order to facilitate the calculation, the rotation axis of the plane set is usually selected as one of the coordinate axes of the coordinate system $\{b\}$. As shown in Fig. 4, we choose the x_b axis as the rotation axis of the plane p_ϕ . When $\phi = 0^\circ$, p_ϕ coincides with the plane $X_b O Y_b$, which means that ϕ is the degree of the rotation from $X_b O Y_b$ to the avoidance plane, and it satisfies the right-hand rule and $-90^\circ < \phi \leq 90^\circ$. In order to obtain the mathematical expression for the boundary of the area which is produced by cutting the collision cone with the avoidance plane, a new coordinate system $\{\Phi\}$ is defined, which is obtained by rotating the coordinate system $\{b\}$ around the x_b axis about ϕ . Let the rotation matrix $R_\phi \in \mathbb{R}^{3 \times 3}$ be defined as

$$R_\phi = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{bmatrix}. \quad (12)$$

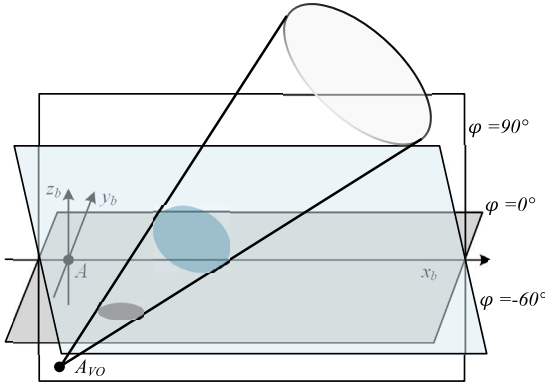


Fig. 4. 3D Absolute Velocity Collision Cone Rotation Tangent.

Then, A_{vo} and B_{vo} can be expressed in the coordinate system $\{\Phi\}$ as

$$A_{vo}^{\Phi} = \begin{bmatrix} x_A^{\Phi} \\ y_A^{\Phi} \\ z_A^{\Phi} \end{bmatrix} = R_{\phi} A_{vo}, \quad (13)$$

$$B_{vo}^{\Phi}(t_p) = \begin{bmatrix} x_B^{\Phi} \\ y_B^{\Phi} \\ z_B^{\Phi} \end{bmatrix} = R_{\phi} B_{vo}(t_p). \quad (14)$$

In order to facilitate the subsequent analysis of the relative position of the avoidance plane and the collision cone, as well as the specific expression of the cross-section, C_{vo} and $G_{vo}(t_p, t_g)$ are represented in the coordinate system $\{\Phi\}$ as

$$C_{vo}^{\Phi} = \begin{bmatrix} x_C^{\Phi} \\ y_C^{\Phi} \\ z_C^{\Phi} \end{bmatrix} = R_{\phi} C_{vo}, \quad (15)$$

$$G_{vo}^{\Phi}(t_p, t_g) = \begin{bmatrix} x_g^{\Phi} \\ y_g^{\Phi} \\ z_g^{\Phi} \end{bmatrix} = R_{\phi} G_{vo}(t_p, t_g). \quad (16)$$

Then, $G_{vo}^{\Phi}(t_p, t_g)$ can be further expressed as

$$G_{vo}^{\Phi}(t_p, t_g) = (B_{vo}^{\Phi}(t_p) - A_{vo}^{\Phi})t_g + A_{vo}^{\Phi} = \begin{bmatrix} (x_B^{\Phi}(t_p) - x_A^{\Phi})t_g + x_A^{\Phi} \\ (y_B^{\Phi}(t_p) - y_A^{\Phi})t_g + y_A^{\Phi} \\ (z_B^{\Phi}(t_p) - z_A^{\Phi})t_g + z_A^{\Phi} \end{bmatrix}. \quad (17)$$

Therefore, in order to obtain the set of all intersection points produced by intersecting the cone surface of the collision cone with the avoidance plane P_{ϕ} , we can set $z_g^{\Phi} = 0$, which is equivalent to $(z_B^{\Phi} - z_A^{\Phi})t_g + z_A^{\Phi} = 0$. Then we have

$$t_g = \frac{z_A^{\Phi}}{z_B^{\Phi} - z_A^{\Phi}}. \quad (18)$$

Therefore, the boundary point set of the collision cone $VO_{A,B}$ on the avoidance plane $X_{\Phi}OY_{\Phi}$ can be expressed as

$$VO_{\phi}(t_p) = \begin{bmatrix} x_{vo}^{\Phi}(t_p) \\ y_{vo}^{\Phi}(t_p) \end{bmatrix} = \begin{bmatrix} (x_B^{\Phi}(t_p) - x_A^{\Phi}) \frac{z_A^{\Phi}}{z_B^{\Phi} - z_A^{\Phi}} + x_A^{\Phi} \\ (y_B^{\Phi}(t_p) - y_A^{\Phi}) \frac{z_A^{\Phi}}{z_B^{\Phi} - z_A^{\Phi}} + y_A^{\Phi} \end{bmatrix}. \quad (19)$$

Where $t_p \in [0, 2\pi]$ and it should meet the following inequalities

$$0 \leq z_B^{\Phi} < z_A^{\Phi} \quad \cup \quad 0 \geq z_B^{\Phi} > z_A^{\Phi} \quad \cup \quad z_B^{\Phi} \cdot z_A^{\Phi} < 0. \quad (20)$$

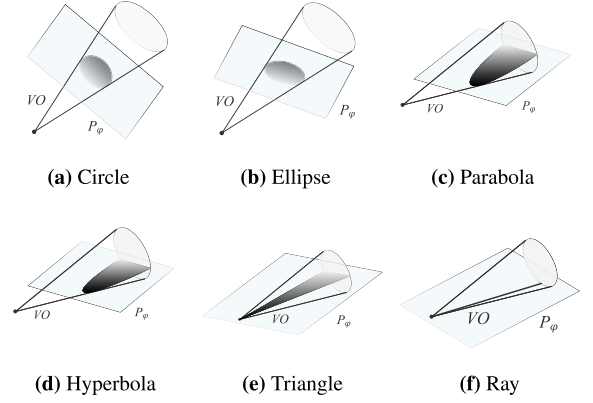


Fig. 5. Several possible cross sections of collision cone cut by avoidance plane.

If (19) has physical meaning, it needs to meet certain restrictions, that is, not every generatrix of the collision cone and its extension line intersect with the plane $X_{\Phi}OY_{\Phi}$. It is necessary to point out that the conditions given in a lot of existing works on three-dimensional velocity obstacle avoidance [22] are $z_B^{\Phi} \neq z_A^{\Phi}$, without considering the case that $t_g < 0$. In this case, the point of intersection of the reverse extension of the generatrix on the collision cone and the avoidance plane is determined. This result has no actual physical meaning for actual planning, therefore, the result obtained by extending the generatrix inversely needs to be eliminated, as shown in (20). Whether there is an effective intersection between the ray and the avoidance plane obtained by the generatrix and its extension depends on the relative position of the obstacle and the UAV, the velocity of the obstacle v_o , and the rotating angle ϕ selected of the avoidance plane. For the convenience of subsequent analysis, θ_{vo} is calculated according to (3), and the pitch angle of the collision cone VO in the coordinate system $\{\Phi\}$ is obtained as

$$\theta_{el}^{\Phi} = \arcsin \frac{z_C^{\Phi} - z_A^{\Phi}}{d_{vo}}. \quad (21)$$

Here, as shown in Fig. 5, according to the relationship between θ_{el}^{Φ} and θ_{vo} , the different position relationships between the avoidance plane and the collision cone are classified and summarized as follows:

- 1) When $|\theta_{el}^{\Phi}| \geq \theta_{vo}$, and $\theta_{el}^{\Phi} \cdot z_A^{\Phi} > 0$, the collision cone does not intersect the avoidance plane.
- 2) When $z_A^{\Phi} = 0$, and $|\theta_{el}^{\Phi}| \geq \theta_{vo}$, the collision cone and the avoidance plane intersect in the cone vertex.
- 3) When $z_A^{\Phi} = 0$, and $|\theta_{el}^{\Phi}| = \theta_{vo}$, as shown in Fig. 5(f), the collision cone is tangent to the avoidance plane on a ray.
- 4) When $z_A^{\Phi} = 0$, and $|\theta_{el}^{\Phi}| < \theta_{vo}$, as shown in Fig. 5(e), the avoidance plane intercepts the collision cone to obtain a triangular-like area, which is the area enclosed by the two rays from the cone point.
- 5) When $-\theta_{vo} < \theta_{el}^{\Phi} < \theta_{vo}$ and $z_A^{\Phi} \neq 0$, as shown in Fig. 5(d), the area obtained by cutting the collision cone with the avoidance plane is one side of the hyperbola and the area it contains.
- 6) When $\theta_{el}^{\Phi} = \theta_{vo}$ and $z_A^{\Phi} < 0$, or when $\theta_{el}^{\Phi} = -\theta_{vo}$ and $z_A^{\Phi} > 0$, as shown in Fig. 5(c), the area obtained by the avoidance plane truncating the collision cone is a parabola and the area it contains.
- 7) When $|\theta_{el}^{\Phi}| > \theta_{vo}$ and $\theta_{el}^{\Phi} \cdot z_A^{\Phi} < 0$, as shown in Fig. 5(b), the area obtained by the avoidance plane truncating the collision cone is an ellipse.
- 8) When $|\theta_{el}^{\Phi}| = 90^\circ$ and $\theta_{el}^{\Phi} \cdot z_A^{\Phi} < 0$, as shown in Fig. 5(a), the area obtained by the avoidance plane truncating the collision cone is a disk.

Different from the two-dimensional plane, for the three-dimensional velocity obstacle avoidance strategy, most of the three-dimensional

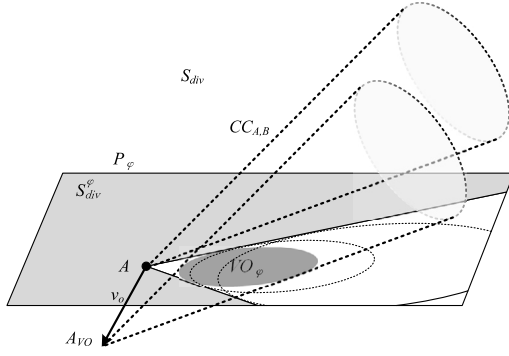


Fig. 6. Divergent velocity domain S_{div} in three-dimensional space and P_ϕ plane.

space belongs to the velocity domain $S_{div} \in \mathbb{R}^3$ which make UAV and obstacle diverge. And the direction of obstacle avoidance is not limited to the front or the rear of the obstacle movement. Therefore, the concepts of S_r and S_f are generally weakened when discussing three-dimensional velocity obstacle avoidance strategies. As shown in Fig. 6, the collision cone $CC_{A,B}$ moves to infinity along the vector v_o . The entire three-dimensional space removes the space passed by the collision cone $CC_{A,B}$ during this process, and the rest is the divergent velocity domain S_{div} in the three-dimensional space. For the P_ϕ plane, during the translation process of the collision cone $CC_{A,B}$, after removing the union of the areas intercepted by P_ϕ , the remaining area is divergent velocity domain S_{div}^ϕ in plane P_ϕ . When $CC_{A,B}$ translates to the end of vector v_o , the area intercepted by plane P_ϕ is VO_ϕ . For the reachable velocity domain, since the quadrotor has high maneuverability and no movement constraints such as turning radius, in order to solve the expected velocity easily, this paper defines the reachable speed domain as a circle area. The center of the circle is the vertical point at the end of the UAV's current velocity vector on P_ϕ , and the radius of the circle can be set within a reasonable range in practical applications.

For a high-real-time system such as a quadrotor, it is very important to calculate the load size. In conventional three-dimensional velocity obstacle avoidance strategies, equiangular spaced avoidance planes are usually set, and the optimal velocity that satisfies the setted performance index is calculated on each avoidance plane, and then the optimal decision-making velocity is obtained by comparing the velocity values of each avoidance plane horizontally. However, such a large amount of calculation might be difficult for the limited onboard computing resource on a small-size UAV. To deal with this issue, we introduce the target point P_{target} and combine it with the selection of the rotating cutting plane's angle ϕ . When the UAV implements obstacle avoidance actions, it considers the translational velocity and direction of the target point, which is practical for the actual application logic, and can reduce the amount of calculation at the same time and improve the real-time performance of the UAVs. Therefore, let the position vector of the target point in the coordinate system $\{b\}$ be defined as $P_{target}^b = [P_x^b, P_y^b, P_z^b]^T \in \mathbb{R}^{3 \times 1}$, the coordinates in the coordinate system $\{\Phi\}$ are defined as

$$P_{target}^\Phi = R_\phi P_{target}^b = \begin{bmatrix} P_x^\Phi \\ P_y^\Phi \\ 0 \end{bmatrix}. \quad (22)$$

Thus, the priority is given to the avoidance plane passing through P_{target}^b or those near that angle. The rotation angle of the P_{target}^b avoidance plane can be obtained as

$$\phi_{target} = \arctan \frac{P_z^b}{P_y^b}. \quad (23)$$

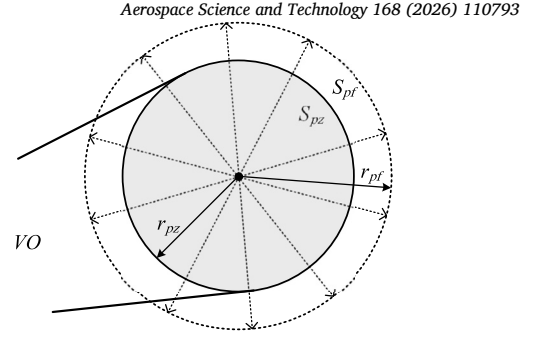


Fig. 7. Combination of spherical potential field and collision cone.

In order to reduce the amount of calculation and avoid the singularity of the boundary of the collision cone section, the parameter t_p is discretized as

$$t_p = \left\{ 0, \frac{2\pi}{n}, \frac{4\pi}{n}, \dots, \frac{2i\pi}{n}, 2\pi \right\}. \quad (24)$$

Among them, n can be flexibly set according to the actual application, and the i -th element of t_p is defined as $t_p(i)$. Therefore, the original VO_ϕ area becomes a polygonal area, and the required performance indicators can be calculated through the polygonal vertex coordinates. Thus the angle ϕ can be chosen by calculating the VO_ϕ of the smallest area for the selected series of avoidance planes. In [23], the selected avoidance plane can be

$$P_{S_{min}} = \{ P_\phi \mid \phi = \arg \min(S_{vo_\phi}) \}. \quad (25)$$

$$S_{vo_\phi} = \frac{1}{2} \left| \sum_{i=1}^n (x_{vo}^\Phi(t_p(i))y_{vo}^\Phi(t_p(i+1)) - x_{vo}^\Phi(t_p(i+1))y_{vo}^\Phi(t_p(i))) \right|. \quad (26)$$

For the selection of the velocity on the selected avoidance plane, this paper selects the included angle with the direction of the target point as the performance index, selects the velocity as far as possible towards the target point, and the z_Φ of the coordinate system $\{\Phi\}$ axis component is 0, as the default representation, the selected velocity is

$$v_{vo}^\Phi = \{ VO_\phi(tp(i)) \mid i = \arg \min(\arccos(\frac{x_{vo}^\Phi(tp(i))P_x^\Phi + y_{vo}^\Phi(tp(i))P_y^\Phi}{\|VO_\phi(tp(i))\| \cdot \|P_{target}^\Phi\|})) \}. \quad (27)$$

2.2. Velocity obstacle avoidance strategy based on the combination of potential field and velocity collision cone

Considering the fact that when the UAV uses the velocity obstacle avoidance strategy to fly to the vicinity of the protection sphere S_{pz} , there may be disturbances in the environment, or there might be control delay caused by the system refresh frequency, or because the obstacle moves too fast. The UAV's position may have fluctuated in S_{pz} or entered its interior. At this time, a singular solution will be generated, which makes it impossible to obtain the expected velocity value normally according to the three-dimensional velocity collision cone, thereby increasing the risk of collision. In order to optimize the velocity calculation logic of the three-dimensional velocity obstacle avoidance strategy, $S_{pf} \in \{(x, y, z) \in \mathbb{R}^3 \mid (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 \leq r_{pf}^2\}$ is defined here as a spherical potential field repulsion domain. As shown in Fig. 7, S_{pf} and S_{pz} are at the same center position, and the radius of S_{pf} is $r_{pf} \in \mathbb{R}^+$, satisfying $r_{pf} > r_{pz}$.

The spherical potential field is a repulsive force field that only exists inside the sphere, and its direction is outward along the radius from the center of the sphere. It is used to generate a radial velocity $v_{pf} \in \mathbb{R}^3$ when the UAV enters the potential field S_{pf} . This radial velocity is conducive to making a quick escape action when external disturbances or obstacles are too fast. It is stipulated here if the UAV enters the protection domain S_{pz} , and it will enter the state of emergency avoidance, and

the expected velocity only comes from the radial velocity given by the potential field domain S_{pf} , which supports the UAV escapes quickly. When the UAV is in the potential field S_{pf} but outside the protection domain S_{pz} , the expected velocity of the UAV is the sum of the radial velocity vector given by the potential field and the one calculated by the three-dimensional velocity obstacle avoidance strategy. This can make the process of the UAV passing near S_{pz} more smooth during the obstacle avoidance procedure, and also avoid the singularity issue. When the UAV is outside the potential field S_{pf} , the potential field will have no effects on the UAV.

The radial velocity v_{pf} provided for the design of the potential field sphere, which is similar as the one in [25], defines the potential field function as follows:

$$Y_{pf}(\lambda) = \int_{r_{pf}}^{\lambda} \Theta_{pf}(x) dx. \quad (28)$$

In (28), the auxiliary function $\Theta_{pf}(\cdot) \in \mathbb{R}$ is defined as

$$\Theta_{pf}(\lambda) = \partial_h\left(\frac{\lambda}{c_{pf}}\right)\Theta(\lambda - r_{pf}), \quad (29)$$

where $\partial_h(\cdot) \in \mathbb{R}^+ \rightarrow [0, 1]$ is the bump function with $h \in (0, 1)$, and is defined as follows:

$$\partial_h(\lambda) = \begin{cases} 1 & \lambda \in [0, h] \\ \frac{1}{2}[\cos(\pi \frac{\lambda-h}{1-h}) + 1] & \lambda \in [h, 1] \\ 0 & \text{other.} \end{cases} \quad (30)$$

The function $\Theta(\cdot)$ is defined as

$$\begin{cases} \Theta(\lambda) = \frac{1}{2}[(a+b)\sigma(\lambda+e) + (a-b)] \\ \sigma(\lambda) = \lambda / \sqrt{1+\lambda^2}. \end{cases} \quad (31)$$

In (31), a , b and e satisfy $0 < a \leq b$ and $e = |a-b|/\sqrt{4ab}$, then $\Theta(0) = 0$. In (31), we can see that when $\lambda \in [0, r_{pf}]$, $\Theta_{pf}(\lambda)$ is a smooth repulsive potential function. When the norm of position vector $P_B^b = [p_{Bx}^b, p_{By}^b, p_{Bz}^b]^T$ of the obstacle B in the coordinate system $\{b\}$ satisfies $\|P_B^b\| \leq r_{pf}$, the radial velocity vector is given by

$$v_{pf} = k_{pf} \Theta_{pf}(\|P_B^b\|_\sigma) \frac{P_B^b}{\|P_B^b\|}. \quad (32)$$

In (32), $k_{pf} > 0$ is a positive gain, $\|\cdot\|_\sigma \in \mathbb{R}^N \rightarrow \mathbb{R}^+$ represents the σ norm which defined as follows:

$$\|\lambda\|_\sigma = \frac{1}{\epsilon} \left[\sqrt{1 + \epsilon \|\lambda\|^2} - 1 \right], \quad (33)$$

and ϵ is a normal number greater than 0. Here we can rewrite v_{vo}^ϕ as $v_{vo}^\phi = [v_{vo x}^\phi, v_{vo y}^\phi, 0]^T$. Let the rotation matrix from the coordinate system $\{\Phi\}$ to the coordinate system $\{b\}$ be defined as

$$R_{(-\phi)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{bmatrix}. \quad (34)$$

Define $v_d \in \mathbb{R}^{3 \times 1}$ as the final selected velocity vector, then we have

$$\begin{cases} v_d = v_{pf} & \text{if } d_{AB} \in [0, r_{pz}] \\ v_d = R_{(-\phi)} v_{vo}^\phi + v_{pf} & \text{if } d_{AB} \in [r_{pz}, r_{pf}] \\ v_d = R_{(-\phi)} v_{vo}^\phi & \text{if } d_{AB} \in [r_{pf}, +\infty]. \end{cases} \quad (35)$$

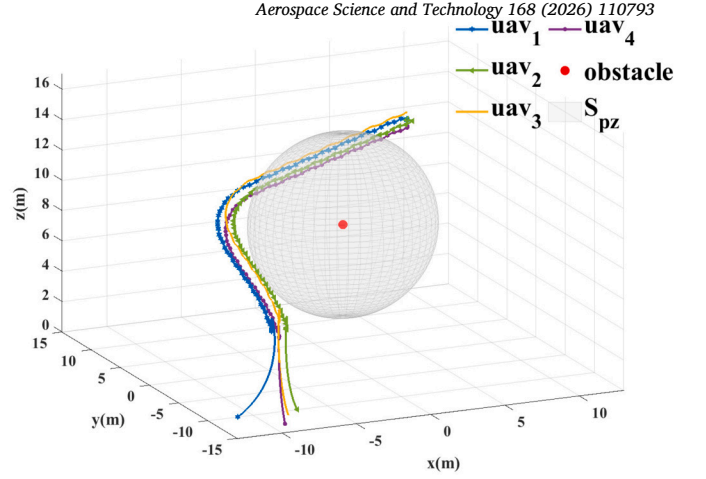


Fig. 8. Numerical simulation: UAV fleet's trajectories when it avoids static obstacle.

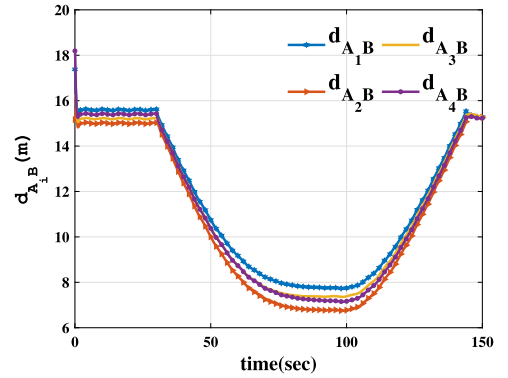


Fig. 9. Numerical simulation: the distance between each UAV and obstacles when the UAV fleet avoids static obstacles.

3. Simulation verification

Considering that it is dangerous to verify the algorithm directly on the quadrotor UAVs testbed, this section firstly conducts two sets of numerical simulations. In this paper, the leader-follower method is used for UAV swarm simulation, where the leader is the virtual expected point, and the trajectory of the virtual center is generated by the three-dimensional velocity obstacle avoidance strategy designed in this paper. The first set of simulations is designed as static obstacle avoidance simulation of the UAV fleet, in which the protection radius r_{pz} of obstacles is designed to be 6 meters, and the potential field radius r_{pf} is designed to be 6.5 meters. In this set of simulation, 4 UAVs take off from the ground, and the target point of the virtual center is chosen as $p_{target} = [9, 12, 12]^T m$. The simulation results are shown in Fig. 8 and Fig. 9.

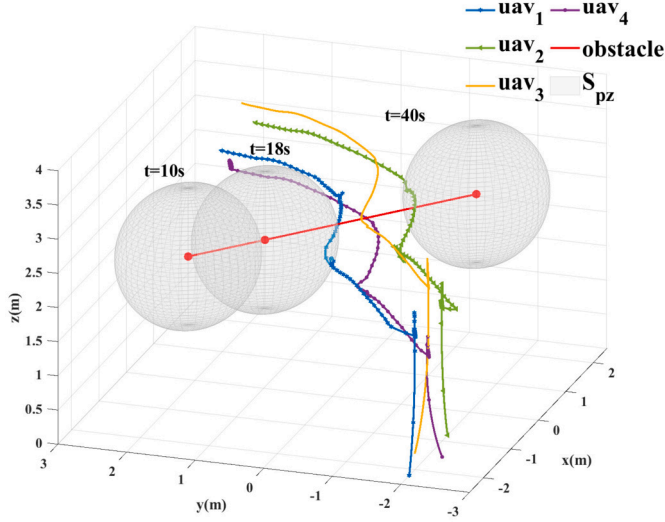
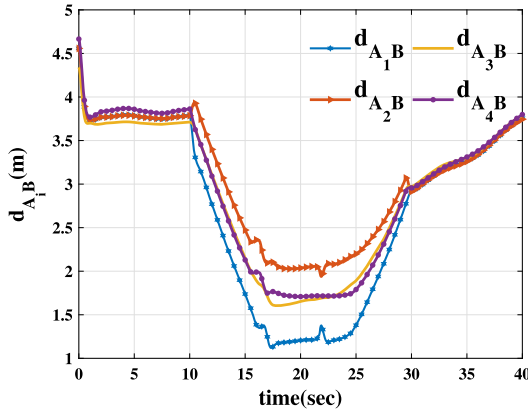
According to Fig. 8, it can be seen that after each UAV takes off from the ground and converge into the desired formation, it can effectively realize the obstacle avoidance function by bypassing the spherical area of obstacles. And it can be seen from Fig. 9 that the distance between each UAV and the obstacle is always greater than the protection radius of the obstacle, indicating that the UAV fleet does not collide with the static obstacle in the numerical simulation.

The second group of simulations is designed as a dynamic obstacle simulation for the UAV fleet, in which the protection radius r_{pz} of obstacles is designed to be 1 meter, and the potential field radius r_{pf} is designed to be 1.5 meters. In this set of numerical simulations, there are also 4 UAVs flying in fleet, and they take off from the ground, converge into formation, and then fly to the target point. The target point of the design virtual center is $p_{target} = [1.5, 2, 3]^T m$, and the obstacle starts from the starting point $p_{obs}(0) = [-1.5, 1.5, 2.5]^T m$ and moves at a con-

Table 1

Comparison of Numerical Simulation Indicators.

Metric	Average	Variance
<i>Length of Path (m)</i>		
Static obstacles	16.075	0.502
Dynamic obstacles	9.050	0.548
<i>Minimum Safety Distance (m)</i>		
Static obstacles	10.837	8.534
Dynamic obstacles	2.934	1.038
<i>Average Computation Time (ms)</i>		
Static obstacles	0.019	2.3×10^{-4}
Dynamic obstacles	0.022	4.1×10^{-4}

**Fig. 10.** Numerical simulation: UAV fleet's trajectories when it avoids dynamic obstacle.**Fig. 11.** Numerical simulation: the distance between each UAV and obstacles during dynamic obstacle avoidance.

stant speed of $v_{obs} = [0.1, -0.1, 0]^T$ m/sec, and the UAV fleet needs to perform the avoidance maneuver to reach the target point.

Fig. 10 indicates the positions of the obstacle protection domains at 10 seconds, 18 seconds, and 40 seconds respectively. It can be seen that the UAV fleet has made corresponding avoidance action during the takeoff and flight to the target point. Each UAV does not collide with the dynamic obstacle while avoiding it, and successfully reaches the target point with a relatively reasonable path. Combined with the Fig. 11, it can be seen that the distance between each UAV and the designed dynamic obstacle is always greater than the radius of the obstacle protection sphere. These results illustrate the validity of the three-dimensional obstacle avoidance strategy proposed in this paper.

Table 1 compares the path length, minimum safety distance, and average computation time between static and dynamic obstacle scenarios. In static scenarios, the longer path and greater minimum safety distance reflect that UAVs can plan more smooth path, taking sufficient distance from fixed obstacles. Dynamic scenarios, however, force more compact flight paths and smaller safety margins. The slight rise in computation time shows the system working harder to handle moving obstacles in real - time. Overall, it proves static scenarios allow pre - planned efficiency, while dynamics demand agile, real - time adjustment, with the system still maintaining basic real - time feasibility.

To validate the effectiveness of our proposed algorithm, we conducted comprehensive comparative simulations. We systematically evaluated the traditional 3D-VO algorithm [26], improved APF algo-

rithm [27], and our proposed method across five obstacle environments of varying complexity, with 10 Monte Carlo simulations performed for each scenario to ensure result reliability.

The simulation results are presented in Table 2. Our results demonstrate that the proposed method achieves significant advantages across all performance metrics. Regarding obstacle avoidance success rate, our method maintains 100% success across all scenarios, while the traditional 3D-VO method completely fails in mixed obstacle environments, achieving only an 80% average success rate. In terms of computational efficiency, our method requires an average computation time of merely 0.02 ms, representing a 99.6% improvement over the traditional 3D-VO's 5.56 ms and a 50% improvement over the improved APF's 0.04 ms, fully satisfying real-time obstacle avoidance requirements.

For path quality assessment, we employ the path length ratio (actual path length to theoretical optimal path length) as our evaluation metric. Our method achieves 0.818, significantly outperforming the traditional 3D-VO's 0.70 (16.8% improvement) and the improved APF's 0.616 (32.8% improvement), generating paths closer to the theoretical optimum. Furthermore, in safety evaluation, our method maintains a moderate safety distance of 0.23 m, effectively avoiding both the overly conservative approach of traditional 3D-VO (1.12 m) that leads to unnecessarily long paths, and the overly aggressive behavior of the improved APF method (which even exhibits -0.03 m collision instances) that compromises safety.

These quantitative results conclusively demonstrate the superiority of our proposed 3D-VO-potential field fusion method in practical applications. All performance improvements pass statistical significance tests ($p \leq 0.01$), and the algorithm exhibits performance variations below 10% across different complexity scenarios, demonstrating excellent robustness and stability.

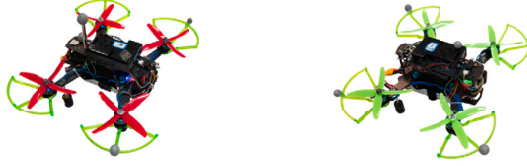
4. Real-time flight experiment verification

In order to further verify the effectiveness of the designed obstacle avoidance strategy, real-time flight experimental verifications are performed on the self-built multi-UAVs flight testbed. However, due to the limited size of the experimental region, it is very difficult to verify the 3D velocity obstacle avoidance of an UAV fleet. Considering the distance $h_i \ll r_{pz}$ between the UAV and the virtual center when the UAV fleet is flying, we use one quadrotor instead of the UAV fleet for obstacle avoidance. Two sets of experiments are performed. In the first set of experiments, the obstacle is set as a static obstacle, and the position is between the starting point and the target point of the quadrotor. In the second set of experiments, the obstacle is set as a dynamic obstacle or a moving one. The obstacle moves in a straight line at a uniform speed in a certain direction, and passes between the starting point and the target point of the quadrotor. If the quadrotor also moves in a straight line at a uniform speed in the direction of the target point, the two will collide with each other. The quadrotor UAVs used in the experiments are shown in Fig. 12, and the screenshot of UAV flight obstacle avoidance experiment is shown in Fig. 13.

Table 2
Overall Performance Comparison.

Metric / Algorithm	Single Static	Single Dynamic	Multiple Static	Mixed	Cluttered	Average
<i>Success Rate (%)</i>						
Traditional 3D-VO	100.0	100.0	100.0	0.0	100.0	80.0
Proposed Method	100.0	100.0	100.0	100.0	100.0	100.0
Improved APF	100.0	100.0	100.0	100.0	100.0	100.0
<i>Average Computation Time (ms)</i>						
Traditional 3D-VO	1.90	2.13	4.95	4.52	14.31	5.56
Proposed Method	0.02	0.03	0.02	0.02	0.02	0.02
Improved APF	0.05	0.07	0.03	0.02	0.03	0.04
<i>Path Length Ratio (Optimal = 1.0)</i>						
Traditional 3D-VO	0.766	0.661	0.775	N/A	0.597	0.70
Proposed Method	0.796	0.795	0.801	0.988	0.711	0.818
Improved APF	0.628	0.552	0.649	0.636	0.616	0.616
<i>Minimum Safety Distance (m)</i>						
Traditional 3D-VO	1.41	0.88	1.32	N/A	0.87	1.12
Proposed Method	0.25	0.21	0.25	0.22	0.22	0.23
Improved APF	0.12	-0.03*	0.13	0.12	0.63	0.19

*Negative value indicates collision with obstacle.



(a) UAV for obstacle avoidance

(b) UAV used as obstacle

Fig. 12. The two UAVs used in the experiments.

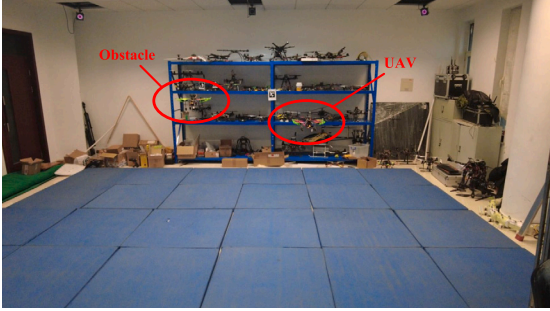


Fig. 13. Screen shot of Three-dimensional velocity obstacle avoidance flight experiment.

The video for the experiments can be found at the following website:
<https://www.youtube.com/watch?v=ZUiMMLpq1M8>.

4.1. Static obstacle avoidance experiment

In this set of experiments, the two quadrotors take off at the same time. The UAV_{obs} which is used as an obstacle will fly to the designed hovering point P_{hover}^I after taking off, and the UAV_{vo} which is designed to perform velocity obstacle avoidance flies to the starting point $P_{vo}^I(0)$ after taking off. After both quadrotors arrive at the starting position, the UAV_{obs} continues to hover at the current position, and the UAV_{vo} will fly to the target point P_{target}^I . The experimental parameters related to this set of experiments are shown in the Table 2.

The experimental results are shown in Fig. 14-16. It can be seen from Fig. 14 that the quadrotor bypasses the designed obstacle protection domain S_{pz} , and flies to the target point obliquely above the starting point with a smoother curve. From Fig. 15-16, it can be seen that the UAV takes off in about 7 – 13 seconds, and then maintains a stable flight at the starting point for about 10 seconds. The UAV completes the entire

Table 3
Parameters in Experiment Case 1.

Symbol	Value
$r_{pz}(m)$	1
$r_{pf}(m)$	1.5
a_{pf}	2
b_{pf}	3
c_{pf}	10
k_{pf}	1
$P_{hover}^I(m)$	$[0, 0, 1.5]^T$
$P_{vo}^I(0)(m)$	$[-1.5, -2, 1]^T$
$P_{target}^I(m)$	$[1.5, 2, 2]^T$

obstacle avoidance action in about 25 – 45 seconds. During this time, the distance between the UAV and the obstacle is constantly approaching. After 45 seconds, the velocity direction of the quadrotor tends to be stable. At this stage, the velocity of the quadrotor pointing to the target point is no longer inside the collision cone. At this point, the distance between the UAV and the obstacle keeps getting farther until the UAV reaches the target point, and the obstacle avoidance procedure ends.

It can be seen from Fig. 16 that the minimum distance between the quadrotor and the obstacle during the whole experiment is about $d_{AB} = 1.31 m$. However, the radius of the spherical potential field is $r_{pf} = 1.5 m$, which means that the potential field provides a certain radial velocity for the UAV during the obstacle avoidance procedure, providing some assurance to prevent the quadrotor from entering the obstacle spherical protection zone. While the minimum distance $d_{AB} = 1.31 m > r_{pz} = 1 m$, the experimental results prove that the designed three-dimensional velocity obstacle avoidance strategy based on the collision cone and potential field can effectively avoid static obstacles, and has achieved good obstacle avoidance performance.

4.2. Dynamic obstacle avoidance experiment

In this set of experiments, the two quadrotors take off at the same time, and the UAV_{obs} acting as an obstacle will fly to the designed starting point $P_{obs}^I(0)$, the UAV_{vo} designed to perform the velocity obstacle avoidance strategy takes off and flies to the starting point $P_{vo}^I(0)$. After both UAVs reach the starting position, UAV_{obs} flies to P_{target}^{obs} with a uniform speed, UAV_{vo} will fly to the target point P_{target}^{vo} . The experimental parameters related to this set of experiments are shown in Tables 3 and 4.

The experimental results are shown in Fig. 17-19. It can be seen from Fig. 17 that the UAV can respond to the designed dynamic obstacle protection zone S_{pz} , and constantly change its own velocity while

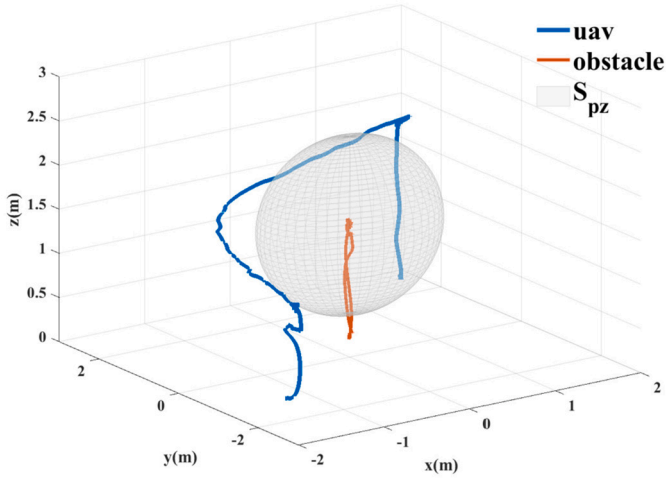


Fig. 14. Experiment Case 1: the trajectory of both quadrotor and obstacle.

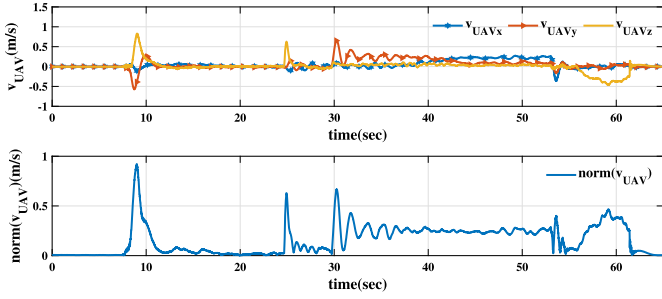


Fig. 15. Experimental Case 1: UAV's velocity curves.

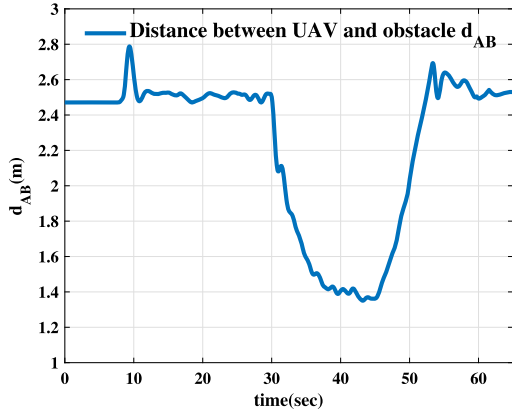


Fig. 16. Experiment Case 1: distance between UAV and obstacle.

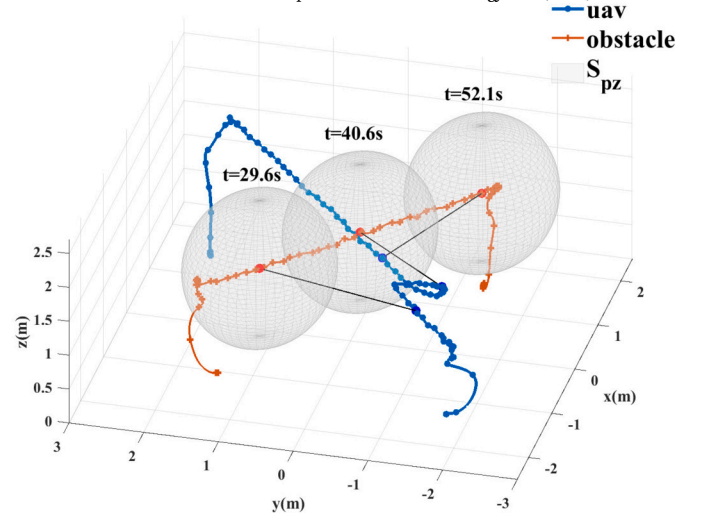


Fig. 17. Experiment Case 2: the trajectory of both quadrotor and obstacle.

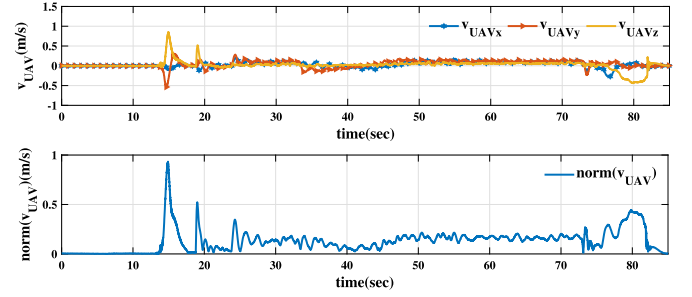


Fig. 18. Experiment Case 2: UAV's velocity curves.

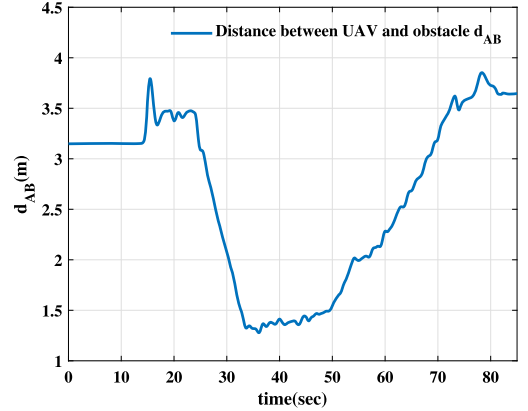


Fig. 19. Experiment Case 2: distance between UAV and obstacle.

Table 4
Parameters in Experiment Case 2.

Symbol	Value
r_{pz} (m)	1
r_{pf} (m)	1.5
a_{pf}	2
b_{pf}	3
c_{pf}	10
k_{pf}	1
$P_{obs}^I(0)$ (m)	$[-1.5, 1.5, 1.5]^T$
$P_{obs}^I(0)$ (m)	$[-1.5, -2, 1]^T$
P_{target}^{vo} (m)	$[1.5, -1.5, 1.5]^T$
P_{target}^{vo} (m)	$[1.5, 2, 2]^T$

maintaining a certain safe distance from the obstacle. The avoidance of obstacles on the trajectory to the target point has been Implemented. Combining Fig. 17 and Fig. 19, it can be seen that the UAV takes off in about 13 to 20 seconds, then starts to fly to the target point, and begins to avoid to obstacles. The UAV completes the entire obstacle avoidance action in about 25 to 55 seconds. During this time, the distance between the UAV and the obstacle is constantly approaching. After 55 seconds, the velocity direction of the quadrotor tends to be stable. At this stage, the velocity of the quadrotor pointing to the target point is no longer inside the collision cone. At this point, the distance between the UAV and the obstacle keeps getting farther until the UAV reaches the target point, and the obstacle avoidance process ends.

Table 5
Comparison of Real-time Experiment Indicators.

Metric	Average	Variance
<i>Length of Path (m)</i>		
Static obstacles	5.203	0.867
Dynamic obstacles	5.836	0.829
<i>Minimum Safety Distance (m)</i>		
Static obstacles	2.197	0.157
Dynamic obstacles	2.735	0.430
<i>Norm of UAV's velocity (m/s)</i>		
Static obstacles	0.2337	0.0477
Dynamic obstacles	0.2673	0.0674
<i>Average Computation Time (ms)</i>		
Static obstacles	4.5	0.67
Dynamic obstacles	5.2	0.26

From Fig. 14, we can see the three key moments of the UAV's dynamic obstacle avoidance. When $t = 29.6$ s, the velocity of the UAV is within the collision cone at this time, and the velocity strategy obtained is to bypass the dynamic obstacle from the upper right, but because of the obstacle's fast speed, the actuator of the UAV has a certain hysteresis, and then the UAV enters the potential field S_{pf} , which generates a radial velocity expectation, which is synthesized with the velocity calculated by the velocity obstacle avoidance strategy. Afterwards, it flies toward the rear right relative to the obstacle. When $t = 40.6$ s, the obstacle is between the target point and UAV's position at this time, the velocity strategy obtained is to fly from the left side of the obstacle protection zone S_{pz} . When $t = 52.1$ s, the dynamic obstacles have been basically avoided, and the quadrotor is heading towards the target point. From Fig. 16, we can see that the minimum distance between the quadrotor and the obstacle during the whole experiment is about $d_{AB} = 1.34$ m. And the set radius of the spherical obstacle potential field is $r_{pf} = 1.5$ m. This means that the potential field provides the radial velocity for the UAV in the procedure of dynamic obstacle avoidance. It brings the assurance to prevent the quadrotor from entering the obstacle spherical protection zone. Since we have the smallest $d_{AB} = 1.34\text{m} > r_{pz} = 1\text{m}$, thus the experimental results prove that the proposed three-dimensional velocity obstacle avoidance strategy based on the collision cone and potential field can also effectively avoid dynamic obstacles, and has achieved good obstacle avoidance performance.

Table 5 compares the path length, minimum safety distance, UAV velocity magnitude, and average computation time between static and dynamic obstacle scenarios in real-world experiments. In static cases, UAVs showed an average path length of 5.203 m and safety distance of 2.735 m, enabling systematic obstacle avoidance with stable velocity and computation time. Dynamic scenarios required longer paths and tighter safety margins, with increased velocity and computation time reflecting real-time adjustments for moving obstacles, verifying the strategy's balance of safety and adaptability.

5. Conclusion

In this paper, for the obstacle avoidance problem of the UAV fleet, an obstacle avoidance strategy based on three-dimensional velocity collision cone and spherical repulsion potential field is proposed. The designed strategy optimizes the avoidance plane selection based on the target point, and reduces the redundancy of calculating the obstacle avoidance velocity on all avoidance planes. This enhances the real-time performance of the UAVs in actual flight. And by taking the singularity issue of three-dimensional velocity obstacle avoidance into consideration, a strategy combining the spherical potential field and the velocity collision cone is designed, which can avoid the singular solution caused by the UAVs fluctuating near the obstacle spherical protection zone. Finally, static and dynamic three-dimensional velocity obstacle avoidance

simulations and real-time flight experiments are performed. Both the numerical simulation results and the real-time flight experimental results show that the three-dimensional velocity obstacle avoidance strategy proposed in this paper has achieved good obstacle avoidance performance, and can effectively avoid both static and dynamic obstacles. Moreover, since the UAVs used in our experiments lacks external perception devices such as stereo cameras, the obstacle safety distances are set overly conservatively. Future work will focus on combining the algorithm proposed in this paper with dynamic sensing to achieve fully autonomous UAV obstacle avoidance.

CRedit authorship contribution statement

Bin Xian: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Conceptualization. **Pengzhi Jiang:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Jinghe Hu:** Writing – review & editing, Writing – original draft, Validation, Methodology, Conceptualization. **Yinxin Wang:** Writing – original draft, Investigation, Data curation, Conceptualization. **Qingsong Li:** Investigation, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Table A.1 summarizes the key variables, constants, and abbreviations referenced in this paper, along with their definitions and contextual descriptions.

Data availability

The authors do not have permission to share data.

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Table A.1

Summary of the key variables, constants, and abbreviations.

Symbol	Definition
UAV	Unmanned Aerial Vehicle
VO	Velocity Obstacle
CC	Collision Cone
APF	Artificial Potential Field
3D-VO	Three-Dimensional Velocity Obstacle
ESDF	Euclidean Signed Distance Field
FASTER	Fast and Safe Trajectory Planner
$P_\phi \in \mathbb{R}^3$	Projection Plane
$S_{pz} \in \{(x, y, z) \in \mathbb{R}^3 (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 \leq r_{pz}^2\}$	Protected Zone Sphere
$S_{pf} \in \{(x, y, z) \in \mathbb{R}^3 (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 \leq r_{pf}^2\}$	Potential Field Sphere
$S_{div} \in \mathbb{R}^3$	Divergent Velocity Domain
$x, y, z \in \mathbb{R}$	Cartesian coordinates in respective systems
$v_q \in \mathbb{R}^{3 \times 1}$	Velocity vector of UAV A
$v_o \in \mathbb{R}^{3 \times 1}$	Velocity vector of obstacle B
$v'_q \in \mathbb{R}^{3 \times 1}$	Relative velocity of UAV A with respect to obstacle B
$d_{AB} \in \mathbb{R}^+ \cup \{0\}$	Distance between UAV A and obstacle B
$r_{pz} \in \mathbb{R}^+$	Radius of the obstacle protected zone
$r_{pf} \in \mathbb{R}^+$	Radius of the spherical potential field
$r_{vo} \in \mathbb{R}^+$	Radius of the circle where collision cone is tangent to S_{pf}
$d_{vo} \in \mathbb{R}^+ \cup \{0\}$	Distance from UAV A to the center of the collision cone
$\theta_{az} \in [0, 2\pi)$	Azimuth angle between x_b axis and AB'
$\theta_{el} \in [-\pi/2, \pi/2]$	Elevation angle between AB axis and $X_b OY_b$ plane
$\theta_{vo} \in (0, \pi/2)$	Angle between AB' and the projection of AB onto the $X_b OY_b$ plane
$k_{pf} \in \mathbb{R}^N \rightarrow \mathbb{R}^+$	Positive gain for potential field radial velocity
$P_{target} \in \mathbb{R}^{3 \times 1}$	Target position vector of UAV
$R_\phi \in \mathbb{R}^{3 \times 3}$	Rotation matrix for avoidance plane P_ϕ around x_b axis
$R_{b_{av}}(-\theta_{el}) \in \mathbb{R}^{3 \times 3}$	Rotation matrix from coordinate system b to c
$c_{v'_q} \in \mathbb{R}^{3 \times 1}$	Relative velocity vector in coordinate system c
$t_g \in [0, +\infty]$	Time parameter for collision cone generatrix
$t_p \in [0, 2\pi]$	Discretized angle parameter for circle B_{vo}
$S_{vo_\phi} \in \mathbb{R}^+ \cup \{0\}$	Area of collision cone section on plane P_ϕ
$v_{pf} \in \mathbb{R}^3$	Radial velocity from potential field repulsion domain
$\Upsilon_{pf} \in \mathbb{R}$	Potential field function for repulsion domain
$\Theta_{pf} \in \mathbb{R}$	Auxiliary function for potential field calculation
$a_{pf} \in \mathbb{R}^+, b_{pf} \in [a_{pf}, +\infty)$	Parameters for potential field function Θ_{pf}
$\partial_h \in \mathbb{R}^+ \rightarrow [0, 1]$	Scaling parameter for potential field domain
$\epsilon \in \mathbb{R}^+$	Positive constant for σ norm calculation
$v_d \in \mathbb{R}^{3 \times 1}$	Final selected velocity vector combining VO and potential field

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